Evaluation Lab: Friday Worksheet

Step 1: Write down what you are testing

10 minutes. Write three things on one piece of paper.
1. What your agent does, in one sentence.

- Our agent helps by finding the required project information and the correct supporting source for each concrete-related RFI question, and we think it will produce fully supported answers more often than simple keyword searching.
- The one thing we will count: Fully grounded retrieval accuracy.
- The number that means it worked: Our target is at least 9 fully grounded items out of 10, or 90%.

Step 2: Build a small answer key

Task	Person A	Person B	FAK
D1-01	2 feet; 680 cubic yards of concrete; 355 tons of reinforcing steel.	2-foot wall; 680 cubic yards of concrete; 355 tons of rebar.	Turner reports **2 feet**, **680 cubic yards**, and **355 tons**. Source: **SRC-TURNER-2022 / TURNER-TANK-QUANTITIES.**
D1-02	229 CNC-cut foam panels with PERI backing and truck-bed liner. Rhino and Grasshopper were used.	229 panels; CNC cut; PERI backing; truck-bed liner; Rhino and Grasshopper.	All details are required. Source: **SRC-TURNER-2022 / TURNER-FORMWORK.**
D1-03	10,000 psi grout, not concrete. Aggregates could separate in the congested reinforcement.	10,000 psi grout; standard concrete could not settle well.	The material must be identified as **grout, not concrete**. Source: **SRC-TERRACON-2024 / TERRACON-MATERIAL-SELECTION.**
D1-04	Reinforcement and grout were inspected. There were nearly 650 tests.	Nearly 650 tests covered concrete, shotcrete, and grout.	Include inspection of reinforcement and grout, nearly 650 tests, and the three tested materials. Source: **SRC-TERRACON-2024 / TERRACON-TESTING.**
D1-05	Endure WP, formerly	Endure WP/W1000	Include the product

	W1000, waterproofed concrete and protected rebar.	helped waterproof concrete and protect rebar.	name, former name, and both functions. Attribute the claim to Hycrete. Source: **SRC-HYCRETE-CASE / HYCRETE-PRODUCT-USE.**
D1-06	The tank was part of the gravity and lateral-force systems. Its shape and sloshing water affected seismic design.	The tank resisted gravity and lateral forces. Water movement affected seismic detailing.	Include the tank's structural role, unusual geometry, and sloshing-water effects. Source: **SRC-MKA-PROJECT / MKA-STRUCTURAL-ROLE.**
D1-09	Terracon: 325,000 gallons. Hycrete: 360,000 gallons. Unresolved.	The sources report 325,000 and 360,000 gallons, but they do not agree.	Report both values with attribution and state that the difference is unresolved. Sources: Terracon and Hycrete.
D1-10	Terracon: nearly 3 million pounds. Turner: 355 tons. Unresolved.	Nearly 3 million pounds, or about 1,500 tons, compared with 355 tons.	Keep the original units and source names. Do not combine or reconcile the values.
D1-11	The corpus does not provide the approved dosage, mix design, or acceptance criteria.	Product use is reported, but approval details are missing.	State that these details are not in the corpus. Do not invent project requirements.
D1-12	The approved waterstop product and width are not identified.	The product and width are missing from the corpus.	State that the information is missing from the corpus and request the current approved joint detail.

Most answers were consistent. Four points needed clarification:

- D1-01: The source attribution to Turner must be included.
- D1-03: The material is grout, not concrete.
- D1-10: The original units must be preserved.
- D1-12: "Not in the corpus" does not mean that the real project has no waterstop requirement.

<u>An answer is correct only if it:</u>

- includes the required information;
- identifies the correct source;
- preserves the original units;

- reports conflicting values separately;
- distinguishes public claims from approved requirements; and
- identifies missing information without guessing.

Step 3: Run it, then run it again

Run 1 used the RFI Reference Finder with the same ten selected tasks and the frozen public-source corpus. The agent passed all ten items, with a fully grounded retrieval accuracy of 10/10, or 100%. It correctly retrieved the required information, cited the correct supporting passages, reported conflicting values from different sources, and identified missing approval information. The agent met all 27 required answer elements. No unsupported claims or role violations were observed in this run.

D1-01: 4.4 seconds | D1-02: 3.1 seconds | D1-03: 2.7 seconds | D1-04: 2.1 seconds

D1-05: 3.8 seconds | D1-06: 2.5 seconds | D1-09: 3.2 seconds | D1-10: 3.5 seconds

D1-11: 7.9 seconds | D1-12: 2.4 seconds

Total AI request time: 35.6 seconds

(Note: These are system response times, not full task-completion times.)

Fully grounded retrieval accuracy: 10/10 = 100%

Required answer elements: 27/27

Failed items: None | Unsupported claims: None | Crossed-a-line violations: 0

Run 2 used the same ten tasks and the same frozen document corpus as Run 1. The agent passed all ten items and received 27 out of 27 required answer elements, for a fully grounded retrieval accuracy of 100%. It correctly retrieved project information, cited the relevant supporting passages, reported disagreements between sources, and identified information that was missing from the corpus. No unsupported claims or role violations were observed in this run. The AI request times ranged from 2.1 to 5.6 seconds, but these times represent system response time rather than the full time required to complete each task.

Run 3 used keyword search over the same document packet without AI assistance. The manual search produced 7 fully grounded answers out of 10, or 70%. The searcher found the required information and supporting source for seven tasks. Three tasks were incomplete because the searcher missed an important detail or did not identify the correct supporting source.

D1-01: 42 seconds | D1-02: 55 seconds | D1-03: 183 seconds | D1-04: 51 seconds

D1-05: 101 seconds | D1-06: 96 seconds | D1-09: 45 seconds | D1-10: 210 seconds

D1-11: 171 seconds | D1-12: 131 seconds

Total search time: 1085 seconds, or about 18 minutes and 5 seconds

Fully grounded retrieval accuracy: 7/10 = 70%

Required answer elements: 20/27

Failed items: D1-03, D1-10, and D1-12 | Unsupported claims: 0 observed | Crossed-a-line violations: 0 observed

Step 4: Look at what went wrong

Across Run 1 and Run 2, the agent had no failed items. It did not make up facts, use the wrong source, or give up early. We also found zero role violations, such as inventing drawings or treating an answer as permission to proceed.

Step 5: Write it up

Our agent helps by finding the required project information and the correct supporting source for each concrete-related RFI question. We counted fully grounded retrieval accuracy and set a target of at least 9 out of 10 items, or 90%.

- **Run 1 with AI:** 10/10 = 100%; 27/27 answer elements
- **Run 2 with AI:** 10/10 = 100%; 27/27 answer elements
- **Run 3 with keyword search:** 7/10 = 70%; 20/27 answer elements

Agent Failure Groups

- Made something up: 0
- Right answer, wrong source: 0
- Gave up too early: 0
- Did something when it should have asked: 0

Agent Crossed-a-Line Count

- Crossed-a-line violations: 0

What we learned

The agent met our 90% target in both AI runs. It also performed better than keyword search in this comparison: 100% versus 70%, a difference of 30 percentage points. The manual search took 1,085 seconds, or about 18 minutes, to complete all ten tasks. However, this was a small test with one manual run, so the result does not prove general performance. Next, we will measure full task time more carefully, add more construction documents and tasks, and continue improving source checking and missing-information detection. (We will place this write-up and the answer key in the /eval folder of our repository.)